Verification round 2: electrical + firmware

Reviewer: independent electrical and embedded-firmware review, 2026-11-02. Repo `/home/user/reading_light`, branch `claude/great-shannon-bikrur` at commit 15d1f96. I did not edit any repo file, and `git status` is still clean.

Result: 0 BLOCKER, 1 MAJOR, 6 MINOR, 13 NOTE. Every round-1 electrical and firmware fix I checked is present in the design files and wired correctly. The new problems come from the fixes themselves:

- Q4 switches hard with no slew control. This is the one MAJOR.
- The "half-current" Green/Blue modes still run at 25 mA peak at level 15.
- The Q4 power-cycle recovery cannot guarantee a TLC59116 power-on reset.
- Some new claims are overstated: TVS → PTC trip, and the PTC hold margin when warm.

Method (what was run or derived, not just read)

Check	Result
Parity: `design.py` pin→net vs a fresh `kicad-cli sch export netlist` vs PCB pad nets (pcbnew). Script: `.../scratchpad/review2/ef_scripts/parity.py`	main-board 105 pins, led-board 72 pins: **0 mismatches**. The generated files match the sources, even though main `design.py` is newer than the schematic (only silk and comments changed)
`OUT_GROUP` derived independently from the exported LED netlist (TLC pin → cathode net → LED ref/die)	`{2,1,0,1,-1,-1,0,2,3,4,1,0,1,0,3,4}`, which equals `board_map.h`
`pio test -e native`	25/25 pass
`pio run` for attiny3216, attiny3216_debug, bench_t_nack, bench_t_stuck. Clean build into a scratch build dir, `-Wall -Wextra`	All succeed. No warnings from `src/` or `lib/core` (only the core's own UARTBAUD `#warning`)
`pytest tests`	4/4 pass
Q4 turn-on inrush: lumped RLC ODE model (battery L/R → Q3/Q1 → C3 → Q2 → C4/C6/C7 → Q4 + cable L/R → LED-board C2). Script: `ef_scripts/q4_inrush.py`	See R2-E1
IREF step chosen by the `tlc_output()` logic (copied verbatim) for every mode and level. Harness: `ef_work/harness/steps.cpp`	See R2-F1
Trust/cap transitions through the real `lightcore.cpp`. Harness: `ef_work/harness/trust.cpp`	See R2-F2
Datasheets fetched this round	Bourns MF-MSMF (`ef_dl/bourns/mf-msmf.txt`). Vishay SMF5V0A, same JEDEC 200 W class as the Littelfuse SMF5.0A, used only as a proxy (`ef_dl/smf/v85881.txt`). Atmel-ICE user guide and MPLAB Snap user guide DS50002787C (`ef_dl/mchp/`). The Littelfuse SMF5.0A itself returns HTTP 403, so it is still **UNVERIFIED**. The Würth RGB is in `ds/`

All paths above are under `/tmp/claude-0/-home-user/7ba8e87c-d4ee-5a66-9891-51bde4e0a9a7/scratchpad/review2/`.

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1. Electrical

ID	Sev	Finding	Evidence	Recommended fix
R2-E1	**MAJOR**	**Q4 switches hard with no slew control, which dips VSYS (the MCU supply) below its guaranteed operating voltage at every LED-board power-up.** PB2 drives the Q4 gate directly: no series R and no gate cap, and Q4 turns on in about 14 ns into 100 ns. VSYS holds about 14 µF (C4, plus C3 behind Q2, plus C6/C7). The LED board holds about 9 µF (C2 near 0 V bias, plus C1/C3). The two share charge faster than the battery path (L ≈ 150 nH, R ≈ 0.25 Ω) can refill it. **Simulated:** VSYS falls to **2.37 V at a 3.7 V cell, 2.17 V at 3.4 V, 2.08 V with an aged cell, and 2.03 V with a low-impedance cable**. It stays below 2.6 V for 2–5 µs. The cable and Q4 see **6–16 A peaks** for about 3 µs; Q4's IDM is 15 A (10 µs). The battery side sees 4–5 A for about 3.5 µs, which should be below typical PCM short-circuit trip levels. This happens at every boot, every resume from empty sleep (SWRR → `setup()` → `led_board_on()`), every recovery power-cycle, and every J3 hot-plug (VERIFY F5). At 10 MHz the ATtiny is guaranteed only down to VBOD, which is 2.4/2.6/2.9 V min/typ/max with a 7 µs typical detection time. So expect either execution out of spec, or a BOD reset. A BOD reset re-runs `setup()`, which switches Q4 on again: at low battery this can become a **boot loop** (about 35 ms period: SUT 32 ms plus init). This problem is new: before round 1, C2 charged through Q2's 1 ms soft start.	`design.py:113-114` (Q4, R12 only; netlist: LED_EN_N = Q4.1, R12.2, U2.9). `main.cpp:192-195` (`pinMode OUTPUT; digitalWriteFast LOW`). DMG2305UX DS p3: tR 13.7 ns, Qg 10.2 nC, Qgd 2.2 nC, IDM −15 A (10 µs). ATtiny3216 Table 36-10: VBOD BODLEVEL2 2.4/2.6/2.9 V, TBOD continuous 7 µs typ. Table 36-3 note 2: 10 MHz "ensured down to ... VBOD with BODLEVEL2". Simulation output in `ef_scripts/q4_inrush.py`; cap DC bias and cable L are **ASSUMED**.	Add gate slew control before the order, which costs two 0603 parts: **10 kΩ in series from PB2 to the gate, and 10–22 nF gate-to-drain** on Q4. R12 stays at 100 k, so Vgs ≈ −0.91·VSYS. Expected ramp ≈ 0.15–0.3 ms, inrush ≈ 0.1–0.3 A, VSYS dip < 0.1 V. Alternatives: an RC like Q2 (10 k + 100 nF gate-source; about 1 A for 0.1 ms, dip about 0.3 V), or 47–100 µF bulk on VSYS, which also addresses round-1 N8. Add a VERIFY item: scope VSYS and U2 VDD at LED-board turn-on with a 3.4 V supply on J2; pass = no dip below 2.9 V and no reset. There is no firmware-only fix (do not pre-charge VLED through SDA/SCL: that back-feeds the slaves' ESD diodes).
R2-E2	MINOR	**The m6 "FIXED" is only partial: the Q4 power-cycle probably does not reset the TLC59116.** A TLC POR needs VCC below 0.2 V. VLED has no bleed path: the pull-ups R8/R9 go to high-Z MCU pins, R2 goes to RESET/OS, the LED anodes go to off sinks, and Q4's body diode points the wrong way. Below about 1 V only µA-level leakage remains, so 10 µF takes seconds to fall from about 1 V to 0.2 V, and `delay(500)` likely leaves VLED at a few hundred mV (estimate, **VERIFY**). The WDT-reset path is off for only about 35 ms. Wiring OS to RESET does nothing for a brown-out, because OS only asserts on temperature. The real brown-out cover is the 1 s MODE1 check, which misses a state that still reads MODE1 = 00h. The LM75B is not affected either way: it needs < 1 V for > 2 s before it resets, and the firmware re-arms it anyway.	TLC59116 DS §9.3.3: "VCC must be lowered below 0.2 V to reset the device"; VPOR 2.5 V (§7.5). LM75B DS §7.9: "below ... ~1.0 V (POR) for over 2 s". `main.cpp:307-313`. Netlist: VLED = J3.2, Q4.3, R8.1, R9.1 only.	Firmware, no hardware change. (1) During the off period, set PB0/PB1 to **OUTPUT LOW**. The bus pull-ups then bleed VLED through 1.65 kΩ (τ ≈ 16 ms, < 0.2 V in about 50 ms). Release the pins before Q4 turns back on. This is safe because the slaves are unpowered, so no diode is forward-biased. (2) Send the TLC **SWRST call** first (address 0x6B, data A5h 5Ah), which the datasheet calls "identical to POR". Update electronics.md "Recovery" and round1.md m6 to "MITIGATED".

ID	Sev	Finding	Evidence	Recommended fix
R2-E3	MINOR	**F1's hold margin is thin in a warm pack, and a nuisance trip latches.** Worst-case normal VBUS draw is about **0.51 A**: 282 mA charge (IREG +10 %), plus about 192 mA LEDs (4000 K full, 177.5 mA nominal, +8 %), 29 mA TLC ICC, about 3 mA MCU, 3 mA charge LED, and 0.5 mA R5. Nominal draw is about 0.46 A. MF-MSMF075 hold current derates to **0.65 A at 40 °C, 0.60 A at 50 °C, 0.55 A at 60 °C and 0.49 A at 70 °C**. F1 sits about 6 mm from U1 (0.36–0.69 W) and D3 (about 0.17 W) in a closed pack, and also heats itself (0.5^2 × 0.11–0.45 Ω). A tripped PTC holds itself tripped, so if it trips, charging stops (the red LED goes off) until USB is unplugged, while the lamp falls back to the battery.	Bourns MF-MSMF DS: MF-MSMF075 Ihold 0.75 / Itrip 1.50 A at 23 °C, Vmax 13.2 V, Rmin 0.11 / R1max 0.45 Ω, 0.2 s at 8 A, Pd 0.8 W; thermal derating table. MCP73831 DS p3: IREG 90/100/110 mA at 10 k (±10 %). TLC59116 §7.5: ±8 %, ICC 29 mA (720 Ω). Placement: F1 (24.0, 47.0), U1 (29.0, 51.0), D3 (29.4, 43.1).	Either add a VERIFY item (F1 case temperature during T3-on-USB plus charging; pass ≤ 50 °C), or move up to **MF-MSMF110/16X** (1.10 A hold, about 0.85 A at 50 °C, 16 V). The cost is a looser cable-short limit on USB (2.2 A trip). Do not use the /8X part (8 V Vmax is too low for the over-voltage case in E4).
R2-E4	MINOR	**"A sustained over-voltage makes the TVS conduct and trips the PTC" is not a sound claim.** An SMF5.0A-class TVS has VBR 6.40–7.1 V at 10 mA and can dissipate only about 0.75 W continuously (RthJA 180 K/W, Tj 175 °C), which is about 0.1 A at the clamp. F1 needs ≥ 1.5 A to trip (less when warm), and at 1.5–3 A that takes seconds; the 0.2 s figure applies only at 8 A. **Band 1, sources ≥ 1.5 A into the TVS:** the TVS dissipates ≥ 10 W for seconds, so it fails, usually short, before or as F1 trips. The result is safe but not resettable. **Band 2, a VBUS of about 6.4–7.5 V:** nothing trips. VSYS = VBUS − Vf(D3) ≈ **5.6–6.6 V**, which exceeds the **ATtiny (6.0 V) and LM75B (6.0 V) absolute maximums** and the TLC's 5.5 V operating maximum. The MCP73831 sits at or above its 6 V operating maximum and near its 7 V absolute maximum.	Vishay SMF5V0A table (same class; Littelfuse **UNVERIFIED**): VBR 6.40–7.1 V, VC 9.2 V at 21.7 A (10/1000 µs), RthJA 180 K/W. Bourns: Itrip 1.5 A, 0.2 s only at 8 A. KEYSPECS: ATtiny VDD abs max 6 V, LM75B VCC 6.0 V, MCP73831 VDD 7 V. electronics.md:18-19; round1.md m3/M-5.	Reword electronics.md and round1.md: "the PTC limits the energy of a gross over-voltage, at the cost of the TVS; a 6.4–7.5 V source is not caught". The accepted residual stays as it is (a USB-C 5 V-only sink with non-compliant chargers excluded). For v0.2, add a real OVP/eFuse on VBUS.
R2-E5	MINOR	**TLC RESET has become a 10 kΩ high-impedance node, and the TLC resets on a 10 ns low pulse.** It used to be tied straight to VCC. Now R2 (10 k) plus the LM75B OS (open drain, up to 10 µA leakage) set the level, on a board where the metal neck passes within about 1–2 mm of the TLC pins (round-1 m-6). A coupled or ESD glitch resets the TLC: the LEDs go dark until the 1 s MODE1 check re-initialises them.	TLC59116 §7.6: tW reset pulse 10 ns min, tRESET 400 ns; §9.3.4: "registers … held in their default states until RESET is again high". LM75B static characteristics (p20): OS ILO 10 µA max, VOL 0.4 V at 3 mA. `led-board/design.py:130-132`.	Add **1–10 nF from RESET_N to GND** at U1 pin 25 (with 10 k that gives τ = 10–100 µs, harmless for the thermostat). Optional: R2 → 4.7 k.

ID	Sev	Finding	Evidence	Recommended fix
R2-E6	NOTE	**J4 VREF through R11 1 k: SerialUPDI and SNAP work, the Atmel-ICE is marginal.** SerialUPDI never uses VREF. SNAP senses VTG, will not run without it, and has VIH = 0.7·VDD logic; its VTG input current is **not specified (UNVERIFIED)**, but a sense input should give < 0.1 V error. The Atmel-ICE draws **< 1 mA from VTG to power its level converters**, which can put up to 1 V across R11. At VSYS 3.3 V its VOH (about 2.3 V) then falls below the ATtiny's VIH of 0.7·VDD (2.31 V). A programmer that supplies power back-feeds at most about 1.3 mA, so the M3 fix holds. Separately, with the handoff's 3.3 V SerialUPDI adapter on USB power (VDD up to about 4.3 V), VIH is 3.0 V, leaving only a 0.1–0.3 V margin.	Atmel-ICE UG: "Draws less than … 1mA for all other interfaces"; "samples the target voltage … to power the level converters". SNAP UG DS50002787C §10.3.1 Table 10-3: VIH = VDD × 0.7; §3: "If the debugger does not sense voltage on its VDD line … it will not operate". ATtiny Table 36-16: VIH 0.7×VDD.	Handoff 04: "use SNAP or SerialUPDI; flash with the board on its battery, not on USB". If the Atmel-ICE is wanted, R11 → 220–330 Ω (sense error ≤ 0.33 V; back-feed still ≤ 6 mA).
R2-E7	NOTE	**USB-unplug handover (R5 10 k):** about 30 ms on Q1's body diode (estimate). VBUS falls quickly to about 3.4 V (charger UVLO plus the charge-LED load), then decays through R5·C1 (τ ≈ 70–80 ms) to the point where Q1 turns on, about VBAT − 0.7 − Vth. During that window VSYS ≈ VBAT − 0.7…0.8 V at 200 mA. With BODLEVEL2 at 2.9 V max, an unplug at VBAT below about 3.65 V under full load can reset the MCU (a visible blink, then recovery). m1 is much better than before, but not closed.	DMG2305UX Fig. 8 (body diode); ATtiny Table 36-10. `design.py:73-74`.	VERIFY E12 already covers this; its fallback "R5 4.7 k" is reasonable (1.1 mA from USB only).
R2-E8	NOTE	**Side effects of F1's drop (0.05–0.23 V at 0.5 A):** (a) On USB, VSYS ≈ 3.9–4.1 V, not the 4.15 V the docs state. (b) On an in-spec weak source (4.75 V at the receptacle, less after the cable), VSYS = VBUS − I·R_F1 − Vf(D3) can sit at about **3.6–3.7 V, at or below `VBAT_RESUME_MV` 3700**. The lamp may then never wake from empty sleep while plugged in, and only wakes after unplugging once the cell has charged. (c) The same low VSYS_RAW forward-biases Q1's body diode at a full cell, so the lamp load is shared from BAT and the MCP73831 never terminates (round-1 M4 residual; the cell floats at 4.20 V).	Bourns R1max 0.45 Ω; `lightcore.h:33`; electronics.md:13.	Document it. Consider `VBAT_RESUME_MV` 3600 with the 3-sample rule, or a VBUS sense pin (m8) in v0.2.
R2-E9	NOTE	The main board has **no VLED test pad**. VERIFY E3/E7 measure VLED and its current "at J3 pin 2". VLED also has no capacitor on the main board (all its capacitance is on the LED board), which is part of why E1 depends on the cable.	Netlist VLED = J3.2, Q4.3, R8.1, R9.1.	Add TP7 on VLED next to Q4.

ID	Sev	Finding	Evidence	Recommended fix
R2-E10	NOTE	F1 uses KiCad's generic `Fuse_1812_4532Metric` (pads 1.125 × 3.4 mm at ±2.1375 mm, gap 3.15 mm). The Bourns MF-MSMF075 is a "Style 1" part (A 4.37–4.73, B 3.07–3.41, D ≥ 0.30 mm), and its recommended land appears to be about 1.5 mm wide with a gap of about 2.7 mm (text extraction ambiguous). The terminations still land on the KiCad pads, but with a smaller heel.	Bourns DS dimension table (MF-MSMF075 → Style 1) and recommended pad layout; footprint file.	Check against the rendered Bourns drawing before release, or make a project footprint.

2. Firmware

ID	Sev	Finding	Evidence	Recommen fix
R2-F1	MINOR	**At their top level, the "half-current" Green and Blue modes still drive 25 mA peaks.** q12 = 2048 gives drive **0.5000**, which is above CG_STEPS[1].cg = **0.4961**, so `tlc_output()` picks **0xFF (25.0 mA) at PWM 128**. Levels 0–13 run at 8.3 mA and level 14 at 12.5 mA, so the M1 mitigation fails exactly at max brightness. LED VF and the TLC knee follow the *peak* current, so Green/Blue at level 15 drop out of regulation below VLED ≈ 3.7–3.9 V, as round-1 M1 described. VERIFY E9 ("Green mode at max … holds 12.4 mA") would read about 12.6 mA *average* on a DMM and pass, which would be misleading.	Harness output: `Green L15: drive G=0.5000 -> step 2 (25.0 mA peak), PWM G=128`. `main.cpp:58-66` (`while (… CG_STEPS[k].cg < mx) k++`). `modes_table.h:20-21`. `gen_modes.py:96-99` (intent: "12.4 mA, IREF step 0xBF"). electronics.md:49-50.	Set G/B to **0 (q12 ≤ 203** `gen_modes.` 2032/4096 = 0.49609 < 0.4961, whic selects 0xBF PWM 255. O the step sea tolerance (c `1.002f < m` Add a host t asserting th Green/Blue modes neve select 0xFF F4). Fix E9 t measure pe current (sco across 1 Ω).

ID	Sev	Finding	Evidence	Recommen fix
R2-F2	MINOR	**Current-cap changes are applied as an unfaded step, and the "frozen" rule fires on old timestamps.** `core_temperature()` writes `c->target` directly through `mode_targets()`, with no `retarget()` and no fade. Harness, 4000 K full: **110.0 mA → 177.5 mA within 1 ms** at the first good reading, about 1 s after every power-on in 3000/4000/6500 K bright levels. The same step happens every time trust toggles. `frozen` measures "time since the reading last changed" whenever the current is above 50 mA, without asking when the load rose. So after a long dim session (constant reading), switching to a bright mode is untrusted *immediately* (capped at 110 mA). Then, 10–60 s later, the first 0.125 °C change jumps it to 177.5 mA. The same 110 ↔ 177.5 mA step can repeat at thermal steady state if the code holds for 5 min.	`lightcore.cpp:149-179` (line 174: `mode_targets(..., c->target)`; line 168: frozen). Harness `trust.cpp`: "1 ms after first good read: 177.5 mA ... step without fade"; "reading changes by 1 LSB: 177.5 mA trusted=1 (1 ms later)".	Call `retarg now)` instea `mode_targe` there, which gives the 16 fade. Start t frozen timer when the cu first exceeds mA, i.e. rese `t_changed_` on that edge tests: no dri step larger t % per ms or trust change "frozen" with min of a loa increase.
R2-F3	NOTE	After the in-loop power-cycle (`main.cpp:309-313`), the next 1 s temperature read can land within the LM75B's first 100 ms conversion. Depending on whether the LM75B actually reset, it returns either a stale value or the power-up register value (0 °C is plausible to the core). A 0 °C reading is "fresh, in range, changed", so it unlocks the 200 mA cap and a thermal factor of 1.0 for one sample. Low impact, because the slew is 10 %/s.	LM75B DS §7.1: "temp reading data is not available until the first conversion is completed in about 100 ms".	Set `last_t` `millis()` (c reads for 15 after `led_board_`

ID	Sev	Finding	Evidence	Recommen fix
R2-F4	NOTE	**The IREF step choice and write ordering live in `main.cpp`, so no host test covers them.** All 25 tests exercise `lightcore` only. A test would have caught F1.	`platformio.ini` `test_build_src = no`. `main.cpp:62-106`.	Move the pu part (step selection, hysteresis, F values, write order) into `lightcore` function tha returns `{coo pwm[16], order}`, and it for every r and level.
R2-F5	NOTE	Nothing ever reads back the LM75B Conf, Tos or Thyst. A corrupted Conf or Tos (bus glitch) silently loosens or disables the hardware thermostat until a TLC failure happens to call `lm75_setup()`. An OS_POL flip would reset the TLC permanently, then trigger power-cycles, then a re-arm: that case heals itself, but a raised Tos does not.	`main.cpp:137-141, 285-287, 302`.	Read Conf (0 and Tos (03h together wit 1 s MODE1 c and re-arm c mismatch.
R2-F6	NOTE	The comments say "the TLC59116 has no bus time-out", which is wrong: it resets its bus interface if SDA or SCL is held low for 25 ms or more. The bus recovery is still worth keeping.	TLC59116 §7.6 note 1. `main.cpp:227`; electronics.md:82.	Fix the text.
R2-F7	NOTE	Handoff 04 says the fuses give "BOD 2.6 V (enabled, sampled in sleep)". The PlatformIO value BODCFG = **0x54** means LVL = 2.6 V, ACTIVE = enabled, **SLEEP = disabled**. That setting is actually the right one for the empty-sleep current. SYSCFG0 0xC5 (UPDI, EESAVE) and OSCCFG 0x02 (20 MHz) are correct.	`~/.platformio/platforms/atmelmegaavr/builder/fuses.py` (`2.6v → 0x54`). ATtiny3216 §6.10.4 BODCFG bit table. `docs/handoffs/04-firmware.md:23`.	Fix the text t "enabled in active/idle, d sleep".

ID	Sev	Finding	Evidence	Recommen fix
R2-F8	NOTE	**Cap arithmetic against what the modes actually draw:** the largest mode is 177.5 mA (4000 K full, nominal; up to about 192 mA at +8 %). The 200 mA "trusted" cap therefore never binds. The 110 mA cap binds in 3000 K (122 mA), 4000 K (177.5 mA) and 6500 K (146.5 mA). The docs quote 182 mA. `I_CAP_UNTRUSTED_MA`'s comment "~0.4 W in the head at 4.2 V" leaves out the TLC quiescent current (about 25–29 mA × 4.2 V ≈ 0.12 W), so the head total at the 110 mA cap is about 0.58 W.	Harness `steps.cpp` "max:" lines. `lightcore.h:22-23`. electronics.md:72.	Correct the numbers. Th thermal revi should confi 110 mA with TLC term included.
R2-F9	NOTE	`lightcore.h:13` points to `docs/design/firmware.md`, which does not exist.	`ls docs/design`.	Fix the refer or add the d

Firmware items checked and found correct

- **CG encoding** (TLC59116 §9.5.9: VG = (1+HC)(1+D/64)/4, CG = VG·3^(CM−1)): 0x7F = 0.3307 (CM=0, 8.3 mA, inside 5–40 mA); 0xBF = 0.4961 (12.5 mA, inside 10–120 mA); 0xFF = 0.9922 (25.0 mA).
- **Write ordering:** gain up writes PWM first, gain down writes IREF first. The in-between state is never brighter than either end.
- **Auto-increment:** AI = 100 is used only for block writes within 00h–1Bh; IREF at 1Ch is written alone with AI = 000.
- **MODE1 and timing:** the MODE1 mask 0x11 matches the POR default (OSC = 1, ALLCALL = 1; AI bits are read-only). The 600 µs oscillator wait covers the 500 µs maximum.
- **LM75B:** Conf 0x08 = queue 2, active low, comparator, normal (Table 8). Tos 60 → 0x3C00 and Thyst 52 → 0x3400 match the 9-bit left-aligned format (Tables 11–13). Temp is read as 11 bits with an arithmetic shift.
- **Watchdog:** WDT 2KCLK = 2.05 s and the PIT = 1 s (same OSCULP32K). The worst loop path, the power-cycle, is about 0.52 s plus I²C timeouts. megaTinyCore 2.6.11 Wire has `TWI_TIMEOUT_ENABLE` by default (`Wire.h:134`), so a stuck bus cannot hang it.
- **Pins and sleep:** megaTinyCore `pinMode()` keeps the ISC bits, so INPUT_DISABLE survives `pinMode(INPUT)` (`wiring_digital.c`). The RTC is enabled before the PIT, and TCD0 drives millis, so there is no RTC conflict.
- **Timer and integer arithmetic:** every timer comparison is elapsed-time and wrap-safe. No integer overflow at real VDD values.
- **Build configuration:** the fault-injection flags exist only in the `bench_*` envs, and `default_envs` is the release env. `upload` does not write fuses (platform `main.py`), so the FW-09 process step is needed.

3. docs/design/electronics.md: statements that are now wrong (R2-D1, NOTE)

Line	Statement	Correction
13	USB + battery: "VSYS ≈ 5.0 − 0.85 = 4.15 V"	With F1, 3.9–4.1 V (E8)

Line	Statement	Correction
18-19	"A sustained over-voltage makes the TVS conduct and trips the PTC"	See E4: either the TVS is sacrificed, or a 6.4–7.5 V source is not caught
26-31	"charger dissipation ≤ ~0.5 W" next to "worst case … ≈ 0.58 W"	The two contradict each other. With IREG +10 % and VPTH 2.79 V the worst case is (5.25 − 2.79) × 0.282 ≈ **0.69 W**
33-34	"worst series drop … 3 × 0.075 Ω"	The LED path now has 4 FETs (Q4) plus J3 and the cable
35-36	Q2 soft start "inrush into ~14–21 µF"	Q2 now charges only about 8 µF. The LED board's 9 µF is switched hard by Q4, which is not covered (E1)
49-50	"Green and Blue modes run at half current (12.4 mA, IREF step 0xBF)"	Only at levels ≤ 14; level 15 runs at 25 mA peak (F1)
54	"8.3 / 12.4 / 24.8 mA"	8.3 / 12.5 / 25.0 mA (the doc's own 25.0 mA full scale)
57	"TLC59116 … 25–37 mA (max)"	≤ 29 mA max for REXT 720–750 Ω; 37 mA is the 180 Ω row (DS §7.5)
60-61	"200 mA total only while the sensor is trusted"	No mode reaches 200 mA (max 177.5 mA), so this cap never acts (F8)
65-66	"after 4 failed inits the firmware power-cycles the LED board"	Not a guaranteed TLC POR (E2)
72	"4000 K at full (182 mA LEDs…)"	177.5 mA nominal
81-82	"the TLC59116 has no bus time-out"	It has a 25 ms time-out (F6)

Also outside electronics.md:

- `docs/verification/VERIFY.md` E9: measure peak current, not average (F1).
- VERIFY.md has no item for the Q4 turn-on dip (E1).
- `docs/handoffs/04-firmware.md:23`: the BOD text (F7).

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4. Round-1 fixes confirmed

Round-1 ID	Confirmed
B1 / FW-01	`board_map.h` matches the LED netlist (derived independently above). `build.py` generates it, and `test_board_map_matches_led_netlist` passes
M2 / M-3 / FW-10 (Q4)	Wiring: G = LED_EN_N (PB2, pin 9), S = VSYS, D = VLED (SOT-23 G/S/D pinout correct). The body diode blocks VSYS → VLED. R12 100 k holds Q4 off during reset, UPDI programming, BOD and WDT resets. Vgs ≈ −VSYS (−2.6…−4.2 V): R_DS(on) ≤ 100 mΩ, about 10–20 mV at 200 mA. Firmware cuts Q4 in empty sleep (`main.cpp:182-190, 204`). **The inrush problem is E1.**
M2 (pull-ups)	R8/R9 go to VLED. No back-feed when the LED board is off: the LM75B has no SDA/SCL→VCC diodes (LM75B DS §8.4: "there are no protection diodes from SCL or SDA to VCC"); the TLC's pins sit at VLED; the MCU pins are inputs with ISC INPUT_DISABLE; PORT OUT stays 0 (TWI0_ClearPins). In sleep the off-state leakage is limited to Q4 I_DSS (≤ 1 µA)
M3	J4.1 = VREF, connected only through R11 1 k (netlist VREF = J4.1, R11.2). The silk and handoff text are updated. A programmer supplying power back-feeds ≤ about 1.3 mA
M4 / m4	R3 3.9 k → 1000/3.9 = 256 mA (±10 %: 230–282 mA), within the 2–67 k range
m1 / m-2	R5 is 10 k. The handover shortens from about 0.3–0.4 s to about 30 ms (E7 residual)
m3 / M-5 / M-6 (F1)	MF-MSMF075-2 sits between J1 VBUS_IN and VBUS/TVS/charger (netlist). Its ratings are confirmed from the Bourns datasheet: 0.75/1.5 A, 13.2 V, 0.11–0.45 Ω, "-2" = tape and reel. **Claims and margins: see E3 and E4**
m6 / M-1 (thermostat)	RESET_N = U1.25 + U2.3 (OS) + R2 10 k to VLED. OS (open drain, active low) is compatible with TLC RESET (VIL 0.3·VCC; OS VOL 0.4 V at 3 mA; tW 10 ns; registers held at default). While OS is asserted the TLC NACKs, `tlc_ok` drops, and the power-cycle is suppressed while the last reading is ≥ Thyst. When OS releases, re-init happens within 0.5 s. **Brown-out coverage: E2. Noise immunity: E5**
B-1 / FW-05	The LM75B is never put in shutdown, is re-configured at every init and TLC retry, and re-armed when it comes back

Round-1 ID	Confirmed
m-3 / FW-04 / F3	`I_CHANNEL_MA` 25.0 = 1.26/750 × 15 × 0.9922. Both the C test and the Python REXT test pass
m-4 / FW-11	Write ordering is correct (see §2)
m-5 / FW-07 / FW-15 / FW-16 / FW-17	Writes are checked. MODE1 is read back every 1 s. Bus recovery (9 clocks plus START/STOP) runs on every LED-board power-up. Reads are sequenced and plausibility-checked; `t` is initialised
FW-02	Blink and fade use elapsed time; the 24.8-day and 49.7-day tests pass
FW-06	The WDT (2 s) is enabled in `setup()` and fed in the loop and on every PIT wake
FW-08	The first ADC conversion is discarded, 8 are averaged, and resume needs 3 × 1 s readings ≥ 3700 mV
FW-09	`fuses` writes BODCFG 0x54 and SYSCFG0 0xC5. The debug build warns when BOD ACTIVE = 0 (mask 0x0C is correct). Handoff wording: F7
FW-12	`RTC.CTRLA = RTCEN` before the PIT (`main.cpp:205-211`). Bench F2/F4 still apply
FW-13	`last_slew` is per instance, and its test passes
FW-18	`-Wall -Wextra` builds clean in project code (all 4 AVR envs rebuilt from scratch)
F9 (CG ranges)	Every step lies inside the DS §9.5.9 ranges (CM=0: 5–40 mA; CM=1: 10–120 mA)
m5 / M-2	Derating is 45/52/58 °C, consistent with Tos 60 / Thyst 52 °C
N9	The LED board `.kicad_pcb` thickness is 1.0 mm
Build gates	ERC 0 on both boards, LED-board DRC 0 errors / 0 unconnected / 0 parity (reports dated after the last regeneration)